



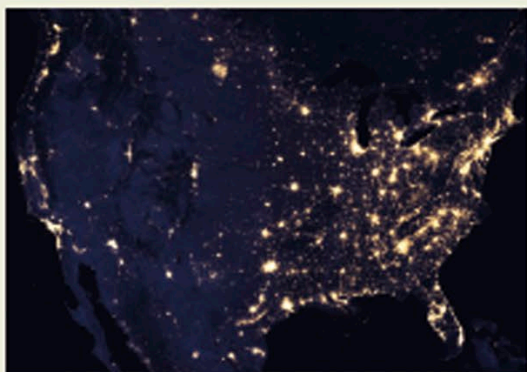
A horse-drawn vehicle shares the street with an electric streetcar in Chicago, 1906.

Electricity takes over

Energy production increased rapidly in the early years of the 20th century. Power lines, streetlights, and neon signs transformed the urban landscape, while trams and electric railroads brought speedier, and cleaner, transportation. Electric goods such as refrigerators made life easier at home.

“Electricity is doing for the distribution of energy what the railroads have done for the distribution of materials.”

American electrical engineer Charles Proteus Steinmetz, 1922



Rivers of light

From the power stations, electricity flows long distances, through high-voltage power lines to substations that supply homes and businesses with energy. This photo, taken from space, shows the cities of the US lit up at night.

Renewable energy

At present, most of the world's electricity is made by burning nonrenewable fuels, such as coal and natural gas, which contribute to global warming (see pp.220–221). Increasing efforts are being made to develop techniques that exploit renewable energy sources.

Solar power

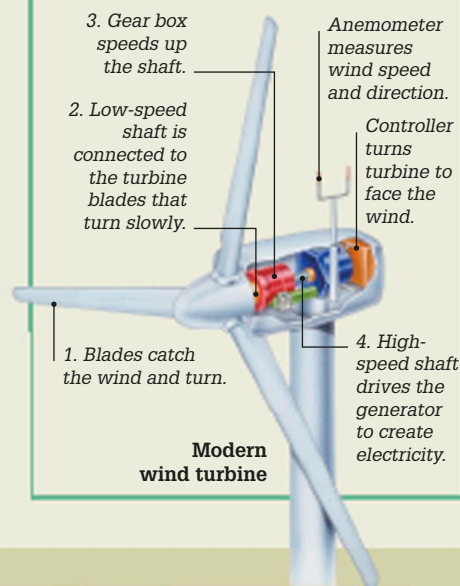
Energy from the Sun is converted into electricity, usually by photovoltaic panels placed in areas receiving direct sunlight. It is a cheap, silent, and nonpolluting source of energy.

Hydroelectricity

The power of falling water is used to turn a turbine to create electricity. It is a reliable energy source, but hydroelectric plants are expensive to build and can affect river flow.

Wind power

Wind is used to turn large turbines to produce electricity. Energy from one turbine can power hundreds of homes.



1884

Charles Parsons, an English engineer, invented the steam turbine, which was used to drive an electric generator, making it possible to produce electricity cheaply and efficiently.

1887

Inventor Nikola Tesla developed an induction motor that ran on AC current, the power delivery system that would later replace the DC system.

1913

The first electric refrigerator for home use came on the market, starting a revolution in domestic appliances.

1954

The world's first nuclear power plant was built to generate electricity for commercial use at Obinsk, Russia.

First electric refrigerator

